

Ruibin Chen

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Brooklyn, NY

EDUCATION

New York University PhD in Electrical and Computer Engineering	Jan 2025 – Present Brooklyn, NY, USA
New York University MS in Computer Engineering • Award: The Myron M. Rosenthal Award for Best MS Academic Achievement	Sep 2022 – May 2024 Brooklyn, NY, USA
University of Nottingham Ningbo China BSc Hons in Computer Science with Artificial Intelligence • First Class Honors • Award: Head's Scholarship (2022)	Sep 2018 – Jul 2022 Ningbo, China

RESEARCH & PROJECT

FR3 Synchronized Channel Sounder Design • Develop a RFSoc 4x2 based FPGA project for FR3 channel sounding. Design and implement the 2 x 2 MIMO channel sounder and "single shot" SISO channel sounding with synchronized header. • Utilize the Vitis and Vivado design tools to implement the BRAM based TX-RX RF system. Design and package the DDR to BRAM bridge Vitis IP and Wave Player IP (generate waveform stream for the RFDC). Configure the Pi-radio "Marie Curie" mixer to convert the 1GHz baseband signal to 10GHz FR3 signal. • Design and implement a Detector IP and Fine-sync IP that detect the candidate incoming signal and precisely synchronized the frame index within the stream beam, providing the ability to estimate the multi-path channel with single-shot measurement. • Data flow: Loading pre-defined IQ waveform from DDR DMA to the BRAM via Vitis bridge IP and generate the stable data stream to the RFDC via waveplayer IP. Control Plane: AXI4-Lite protocol. Clock Domain: PS low speed at 100MHz, DDR 300MHz, and PL 245.76 MHz clock domain.	July 2026 – Present
TurtleBot-based FR3 TX localization Lead programmer • Managed the network configuration among the TurtleBot, visualization center, and FR3 receiver system, and bring up the FastDDS system for the ROS2 framework. Programmed the interface for turtlebot moving and PTUD48 Gimbal control. • Encapsulated the ROS2 topic publisher interface for the FR3 system to publish the Angle of Arrival to the ROS2 network. Design the slam visualization RViz add-on marker arrays to draw arrows for body orientation & AoAs, and the history lines to guess the Tx location at the line intersection points.	Oct 2025 – Sep 2026 Brooklyn, NY, USA
Multi-Cell Multi-Band Handset Digital Twin for Learning-Based Closed-Loop Array Activation Research Assistant & First Author • Developed a Sionna ray-tracing-based UE digital twin combining NYU Tandon 3D urban geometry, multi-cell FR1/FR3 base stations, and an embodied eight-array handset with band-dependent antenna patterns. • Generated time-resolved CQI and rate traces under stochastic pedestrian mobility and handset self-rotation, capturing multi-band channel dynamics across feasible cell-array links. • Formulated an energy-constrained closed-loop array-activation problem and designed PPO-based rate-aware/risk-aware policies, achieving stronger best-rate performance than greedy and bandit baselines.	Sep 2025 – May 2026 Brooklyn, NY, USA
UAV RCS simulation for Integrated Sensing and Communications Research Assistant • Implemented the UAV RCS channel simulation algorithm in NVIDIA Sionna and the UAV 3D random flying trajectory algorithm. Generating datasets for Integrated Sensing and Communications	Feb 2026 – Aug 2026 Brooklyn, NY, USA
Transformer-Based Rate Prediction for Multi-Band Cellular Handsets Research Assistant & First Author • Developed a Sionna ray-tracing-based handset mobility simulator with a metal body model featuring dual FR3 (15 GHz) and FR1 (3.5 GHz) antennas, enabling UE translation and self-rotation on the NYU Tandon map to compute per-antenna channel capacity. • Formulated a multi-band channel rate prediction problem for frequency hopping under energy constraints, targeting accurate per-antenna rate prediction with sparse historical measurements.	Feb 2025 – Sep 2025 Brooklyn, NY, USA

- Designed a transformer-based neural predictor combining temporal convolutional encoding and cross-antenna self-attention, achieving $\approx 28\%$ lower MSE and improved robustness under high-mobility fading for energy-efficient band selection.

Interpolation Techniques for Fast Channel Estimation in Ray Tracing

Mar 2024 – Aug 2024

Research Assistant & First Author

Brooklyn, NY, USA

- Proposed a reflection-model-based interpolation framework to accelerate ray-traced MIMO channel estimation. Implemented path clustering and kernel regression on reflected image sources to achieve high-fidelity spatial interpolation.
- Conducted Sionna simulations in urban LOS/NLOS scenarios at 28 GHz, validating total received power and capacity accuracy across 1–8 m grids for wide aperture MIMO Transmitter Antenna Arrays.

Efficient and Plug-in Named Data Network (NDN) in Health Monitor System

Jan 2021 – Oct 2021

Research Assistant & Contribution Author

Ningbo, China

- Proposed the Neighborhood Registration Scheme to address producer mobility, reducing packet loss by over 80% and delay by 40% versus prior methods. Implemented and evaluated routing algorithms using ndnSIM (ns-3) and MATLAB, independently conducting experiments and analysis under faculty supervision.

Efficient Wireless Sensor Network Architecture Design

Jun 2020 – Aug 2020

Research Assistant & First Author

Ningbo, China

- Proposed an optimized clustering scheme by considering communication distance and residual energy. Selected weighted energy-efficient cluster heads to improve the overall lifetime of the Wireless Sensor Network. Completed simulation in MATLAB.

Breath Monitoring System Design

Nov 2022 – Dec 2022

Team Leader & Algorithm Engineer

New York City, USA

- Developed a hardware device for breath monitoring that detects asphyxia using CO₂ concentration and alerts users via LCD and LED indicators. Implemented an event-driven system on an STM32F4 platform with an SGP30 sensor using PlatformIO, and led a four-member team to successfully deliver the project.

PUBLICATION

- Lei, H., Chen, R., Jiang, Y., Rasteh, A., Dhananjay, A. and Rangan, S., 2026. MAGNETAR: Multipath-Guided Spatial Posteriors for Transmitter Pose Inference in the Upper Mid-Band. *arXiv preprint arXiv:2609.20670*.
- Chen, R., Lei, H., Guo, H., Mezzavilla, M., Poddar, H., Yoshimura, T. and Rangan, S., 2026, May. Transformer-based rate prediction for multi-band cellular handsets. In *2026 IEEE International Conference on Communications Workshops (ICC Workshops)* (pp. 1-6). IEEE.
- Chen, R., Joy, J., Hu, Y., Yin, M., Mezzavilla, M. and Rangan, S., 2024, October. Interpolation Techniques for Fast Channel Estimation in Ray Tracing. In *2024 58th Asilomar Conference on Signals, Systems, and Computers* (pp. 1383-1388). IEEE.
- Kar, Pushpendu, Ruibin Chen, and Yukai Qian. "An efficient producer mobility management technique for real-time communication in NDN-based remote health monitoring systems." *Smart Health* 26 (2022): 100309.
- Chen, Ruibin, et al. "WLEACH-CK: Weighted K-means based LEACH-C algorithm for cluster head selection." *2021 17th International Conference on the Design of Reliable Communication Networks (DRCN)*. IEEE, 2021.

INTERNSHIP

China Telecom

Oct 2024 - Dec 2024

Network Operations Intern

Shaoxing, China

- Analyzed cellular network performance using RSRP to assess coverage, handover behavior, and channel quality, supporting network optimization and coverage improvements

SKILLS & INTERESTS

- Programming Skills: Python (7 yrs), C (4 yrs), C++ (4 yrs), MATLAB (4 yrs), Java, OpenGL, OpenCV, Git, JavaFX
- Simulation & Framework: NVIDIA Sionna, ROS2, Remcom Wireless Insite
- Robotics & Platform: Turtlebot4, PTUD48 Gimbal